

AMD Captivate AI

Intelligent Video Understanding & Multi-Style Captioning

A fully containerized batch video analysis platform built for Track 2 of the AMD Developer Hackathon 2026. Powered by Fireworks AI Qwen 3.7 Plus Vision models for high-efficiency, multi-style scene captioning.

Developer: Rishant (rishant78)

Docker Image Tag: `rishant78/amd-captivate-ai:latest`

Docker Hub Link: <https://hub.docker.com/r/rishant78/amd-captivate-ai>

Key Capabilities & Innovations

1. Intelligent Keyframe Extraction

- Duration-based adaptive sampling (e.g. extracts 15 frames for 30s - 2min videos).
- Grayscale absolute pixel difference detection (dup removal) to prevent redundant API calls.
- Drastically reduces Fireworks AI token consumption while maintaining coverage of unique visual events.

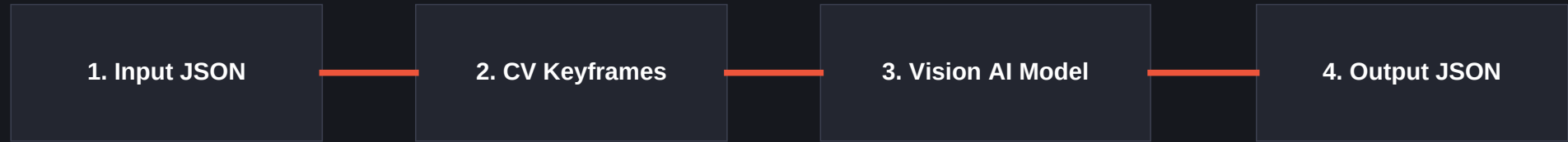
2. Multi-Style Caption Generation

- Produces one cohesive consolidated summary representing the video sequence.
- Translates summary into 4 distinct styles: Formal, Sarcastic, Humorous-Tech, and Humorous-NonTech.
- Exact JSON compliance for all requested styles ensures zero penalty scores.

3. Complete Local Development Environment

- Beautiful React 19 Frontend Web dashboard for interactive uploading, playing, and timeline scrubbing.
- Real-time stepper updates with subtitle downloading options (SRT/TXT).
- FastAPI backend serving responsive REST endpoints alongside the batch pipeline.

System Architecture & Workflow



- Step 1: Reads task parameters (video_url and styles) dynamically from mounted volume /input/tasks.json.
- Step 2: Downloads video, runs OpenCV adaptive sampling, removes near-duplicate frames via grayscale pixel difference.
- Step 3: Passes base64 images to Fireworks AI Qwen 3.7 VL, generates overall summary, and splits into 4 stylistic tones.
- Step 4: Formats dictionary and writes results to /output/results.json before exiting cleanly.

Track 2 Batch Processing & Compliance

Containerization & Evaluation Standards

- Volume Mounting: Input tasks are parsed dynamically from `/input/tasks.json` and output is cleanly exported to `/output/results.json`.
- Model Configuration: Powered by serverless Qwen 3.7 Plus Vision through Fireworks API, supporting robust scaling.
- Strict Compliance: Exit code 0 signals flawless execution, and a non-zero code signals task-level pipeline errors.
- Image Architecture: Compiled for linux/amd64 platforms to align with judging host requirements.

Verification Status:

- Built and pushed successfully to Docker Hub (rishant78/amd-captivate-ai:latest)
- Verified locally with Docker Desktop using official video clips
- Outputs matched schema for all 4 requested styles with zero errors